



# Grade 5 Mathematics

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADE 5 · FULL YEAR

~12½ hrs per week · 150 min/day, M–F

**About this overview.** This document is an example of how the course might be taught. Scholars may see some variation in lesson order, activities, and examples from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Grade 5 Mathematics follows the Dimensions (Singapore-style) approach: every new idea is met first with hands-on or visual models, then pictures, then numbers and equations. Scholars complete the elementary arithmetic story — all four operations with fractions and decimals — then step toward middle school with expressions, volume, ratio, rate, and percentage. Bar models remain the workhorse for multi-step word problems. The year runs as 15 units, each with daily lessons, a mid-unit Concept Check on longer units, and a Summative Assessment at the end.

## THE YEAR AT A GLANCE

### 1 Whole Numbers and Expressions (Units 1–3)

Place value, writing and evaluating expressions, and multi-digit multiplication and division.

### 3 Measurement and Volume (Units 7–8)

Measurement conversions and the volume of solid figures.

### 5 Geometry and Data (Units 11–12)

Two-dimensional figures and the coordinate plane; data analysis and graphs.

### 2 Fractions (Units 4–6)

Adding, subtracting, multiplying, and dividing fractions with understanding, not just rules.

### 4 Decimals (Units 9–10)

Decimal place value and all four operations with decimals.

### 6 Ratio, Rate, and Percentage (Units 13–15)

A first look at proportional reasoning, heading into middle school mathematics.

## A TYPICAL WEEK

- A new lesson most days, built in four parts: Warm-Up, Learn, Practice, and Show What You Know.
- Concrete-pictorial-abstract teaching: hands-on or visual models first, then pictures, then numbers and equations.
- Bar models for word problems and carefully varied practice so each skill becomes automatic.
- A Concept Check halfway through longer units and a Summative Assessment closing every unit.

## TIME COMMITMENT

### 750 minutes / week

≈ 2½ hours a day, Monday–Friday

- The daily lesson and its practice fill most of the scheduled math time.
- Concept Check and Summative Assessment days replace new lessons as each unit closes.

## WHAT SCHOLARS NEED

- A dedicated math notebook, pencil, and paper — every problem gets worked by hand.
- Simple counters or everyday objects for the hands-on step of new concepts.
- A computer with reliable internet and access to Canvas.

**On-Demand (asynchronous).** Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

**Live school.** Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting daily as a core course.